



Time-delayed generalized Korteweg–de Vries–Burgers equation: Well-posedness and exponential decay

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ABSTRACT

This study explores the generalized Korteweg–de Vries–Burgers equation incorporating delay feedback and a damping term. Using semigroup arguments and Lyapunov functional techniques, we establish the existence of a global solution when the exponent of the nonlinear term satisfies some growth conditions. Furthermore, we prove exponential stability estimates under suitable assumptions: first in the case of a positive damping coefficient, then within a more comprehensive framework, accommodating sign changes in both coefficients, i.e. for the damping and the delay feedback. In both cases, we adopt refined conditions on the delay feedback's coefficient, extending and enhancing existing results in the literature. In particular, our conditions are independent of the time delay size.

1. Introduction

In 1895, Korteweg and de Vries [30] formulated the Korteweg–de Vries (KdV) equation to describe weakly nonlinear waves. This equation has been widely applied across various disciplines to model significant physical phenomena. A related extension, the Korteweg–de Vries–Burgers (KdV–B) equation, was introduced by Su and Gardner [38] to investigate the combined effects of dispersion, dissipation, and nonlinearity in wave propagation through a liquid-filled elastic tube [23].

The KdV–B equation integrates features of the KdV equation and the Burgers equation, making it a valuable tool for describing nonlinear wave propagation in dissipative and dispersive media. Its standard form is given as follows:

$$u_t + uu_x - \nu u_{xx} + \mu u_{xxx} = 0, \quad (1.1)$$

where:

- $u(x, t)$ is the dependent variable representing the wave amplitude or profile,
- $\nu > 0$ is the viscosity (dissipation) coefficient, representing dissipative effects,
- $\mu > 0$ is the dispersion coefficient, characterizing wave spreading,
- $t > 0$ and $x \in \mathbb{R}$.

If $\nu = 0$, the equation reduces to the Korteweg–de Vries (KdV) equation, while for $\mu = 0$, it corresponds to the Burgers equation. The KdV–Burgers (KdV–B) equation finds extensive applications in physics and engineering. In fluid dynamics, it models shallow water waves under the influence of viscosity [18]. In plasma physics, the equation describes ion-acoustic waves, capturing the interplay between weak collisional effects and dispersive dynamics [22]. The KdV–B equation is instrumental in analyzing soliton behavior,

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wave damping, and the transition between dispersive and dissipative regimes. Beyond these, its versatility extends to modeling density waves in traffic flow, light pulse propagation in lossy nonlinear optical media, and geophysical flows, including atmospheric gravity waves and oceanic phenomena near the seabed. We refer also to the book [40] for a physical deduction of the KdV equation.

The literature has shown considerable interest in the stability of KdV equations. The work in [41] presented one of the first works on the exponential stabilization of the KdV equation on a bounded domain. Stabilization is accomplished under an appropriate supposition regarding the feedback's weight in this study, with a spatial domain length of $L = 1$. The exponential stability of the nonlinear KdV equation can be achieved without the need for a boundary feedback law, as shown in [32], unlike the approach in [41], for the case of noncritical lengths. Nevertheless, it has been shown in [32] and [35] that the addition of a localized damping term to the nonlinear KdV equation allows to achieve exponential stability also for critical lengths. These works further establish the unique continuation property for weak solutions. Furthermore, [15] and [39] have established asymptotic stability results for the nonlinear KdV equation for specific critical lengths.

In [3], the authors proved that the solution of (1.1) satisfies the following decay estimate:

$$\|u(\cdot, t)\|_{L^2(\mathbb{R})} \leq C t^{-\frac{1}{4}},$$

for all $t > 0$ and some positive constant C , establishing that this decay is polynomial. In [12], the authors considered the damped Korteweg-de Vries type equation:

$$\begin{cases} u_t + bu_x + u_{xxx} + uu_x + au = 0, & (x, t) \in \mathbb{R} \times (0, \infty), \\ u(x, 0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad (1.2)$$

where b is a real constant and $a = a(x)$ is a non-negative function. They specifically studied the case where $a(x) \neq 0$ and satisfies $a(x) \geq \alpha_0$, with $\alpha_0 > 0$ being a constant. For related results, we refer the reader to studies that consider periodic conditions [26] and more general nonlinearities [37]. Subsequently, in [11], the authors investigated a more general case by studying the KdV-B equation, given as:

$$\begin{cases} u_t + u_{xxx} - u_{xx} + uu_x + au = 0, & (x, t) \in \mathbb{R} \times (0, \infty), \\ u(x, 0) = u_0(x), & x \in \mathbb{R}. \end{cases} \quad (1.3)$$

Here, the function $a(x)$ is allowed to change its sign. The authors proved that if $a \in L^\infty(\mathbb{R})$ and there exist a positive constant a_0 and a function $a_1 \in L^p(\mathbb{R})$ for some $1 \leq p < \infty$, such that the following assumptions are satisfied:

$$a \geq a_0 + a_1 \quad \text{almost everywhere and} \quad \|a_1\|_{L^p(\mathbb{R})} \leq \left(\frac{a_0}{c(p)} \right)^{1-\frac{1}{2p}},$$

for some constant $c(p)$ depending on p , then the problem (1.3) is well-posed and exhibits exponential stability. The KdV-B equation has been extensively studied in the literature, with key contributions from works such as [5,7,25]. Additionally, we highlight [19], where the authors examined the well-posedness and exponential stabilization of a generalized Korteweg-de Vries-Burgers equation posed on the whole real line, under the influence of a damping term. Both well-posedness and stabilization results were established for cases when the nonlinear term satisfies some growth conditions. They also provided global well-posedness results in $H^s(\mathbb{R})$ for certain values of s . We mention also [27] for a boundary controllability result of a Korteweg-de Vries-type Boussinesq system.

Incorporating time delays into partial differential equations is a topic of significant interest in both mathematics and applied sciences. Time delays account for the dependence of system dynamics on past states, a phenomenon commonly observed in physical, biological, and engineering systems. For instance, in control systems delays arise naturally due to finite signal transmission speeds, while in population dynamics they can model processes such as maturation or gestation periods.

When a delay is introduced into a PDE, such as the Korteweg-de Vries-Burgers (KdV-B) equation, it can significantly affect the system's dynamics. For models that are uniformly asymptotically stable, the addition of even a small time delay term may lead to phenomena such as destabilization or oscillatory modes, as has been observed in control systems (e.g., [16,17]). Stability analysis in the presence of time delay often requires advanced techniques, including spectral analysis and the use of Lyapunov-Krasovskii functionals.

However, the appropriate choice of time delay can restore stability [21], and this can also be achieved through suitable feedback laws [2,31,34]. For the analysis of evolutionary partial differential models in presence of time delay effects see also [1,4,8,13,20]. For recent developments a KdV-type equations with delays we refer to [9,10].

The study of time-delayed KdV-Burgers models is not only mathematically rich but also closely connected to real-world applications where dispersive-dissipative dynamics interact with delay effects. In traffic flow theory, for example, driver reaction times and memory effects have been shown to lead naturally to delayed KdV-Burgers formulations that capture the emergence and dissipation of stop-and-go waves in congested regimes [24]. In engineering, delayed feedback control of dispersive waves arises in distributed systems subject to actuation or sensor lags, and stabilization strategies for KdV-Burgers equations with uncertain time delays have been successfully developed using Lyapunov-Krasovskii techniques and linear matrix inequalities [25]. From a theoretical standpoint, [29] rigorously established the well-posedness and exponential decay of KdV-Burgers systems under time-delayed feedback, even allowing indefinite damping terms. Overall, these studies underscore that the class of problems addressed here is not merely abstract but has concrete implications for the modeling of traffic, plasma dynamics, viscoelastic flows, and delay-robust control design.

In this paper, we consider the generalized KdV-B equation and aim to relax the conditions on the time delay and damping term compared to those found in existing literature, such that exponential stability is achieved. Notably, the results presented here cannot be deduced from the general approaches proposed in [28,34]. A KdV model on a finite interval with boundary time-delay feedback

was recently studied in [6], where the authors established well-posedness and exponential stability. Similarly, [29] analyzed the well-posedness and exponential stability of the KdV-B model under specific assumptions on the damping and delay terms.

Additionally, in the recent work [14], the well-posedness and boundary stabilization of the KdV-B equation were studied, with boundary control of memory type. In this paper, we investigate the time-delayed generalized Korteweg-de Vries-Burgers equation with a damping term $h = h(x)$, where $h(x)$ is allowed to change sign. The system under consideration is as follows:

$$\begin{cases} u_t(t, x) + u_{xxx}(t, x) - u_{xx}(t, x) + h(x)u(t, x) + \lambda(x)u(t - \tau, x) + a(u)u_x(t, x) = 0 & \text{in } \mathbb{R} \times (0, \infty), \\ u(x, s) = u_0(x, s) & \text{in } \mathbb{R} \times [-\tau, 0], \end{cases} \quad (1.4)$$

where the coefficients $h(x), \lambda(x) \in L^\infty(\mathbb{R})$ and the constant $\tau > 0$ is the time delay. Moreover, we assume that $a \in C^1(\mathbb{R})$ is a positive real-valued function and at some point we need it to satisfy the following growth conditions

$$|a(\mu)| \leq C(1 + |\mu|^r) \quad \text{and} \quad |a'(\mu)| \leq C(1 + |\mu|^{r-1}), \quad \forall \mu \in \mathbb{R}, \quad (1.5)$$

for some $C > 0$ and $1 \leq r < 2$.

We observe that under the assumption $h(x) \geq \alpha_0$ for almost every $x \in \mathbb{R}$, if the coefficient of the delay term λ satisfies $\|\lambda\|_{L^\infty(\mathbb{R})} \leq \alpha_0$, exponential decay can be derived straightforwardly, as the delay effect can be counterbalanced by the undelayed damping term as in [33]. This paper extends the analysis to more general scenarios. One such case considers h as being bounded below by a positive constant while allowing $|\lambda(x)| \geq h(x)$ in certain regions of the domain. Additionally, we explore situations where the coefficient of the undelayed feedback is indefinite, namely it can change sign. The main contributions of this paper are twofold:

- The analysis of the well-posedness and the stability of the generalized Korteweg-de Vries (KdV-B) equation incorporating a time delay and a generalized nonlinearity, which has not been previously explored in the literature.
- A significant enhancement of the theoretical framework by relaxing the assumptions required to establish stability. In particular, we are able to obtain exponential stability under conditions independent of the time delay (see, for instance, (2.3)–(2.8) below). On the contrary, the conditions considered in the recent literature require a bound on the time delay size (see Remark 2.3 below).

The paper is organized as follows: Section 2 discusses the well-posedness and exponential stability of the linear model under enhanced assumptions on the delay term. Section 3 introduces the nonlinear model and presents results on its well-posedness and exponential decay. Finally, Section 4 addresses a more general setting where the undelayed damping coefficient is allowed to change sign too.

To streamline notation throughout this paper, we use $\|\cdot\|_p$ and $\|\cdot\|_\infty$ to represent the standard norms on $L^p(\mathbb{R})$ and $L^\infty(\mathbb{R})$, respectively. We also denote by C a generic positive constant, whose value may vary from line to line.

2. Well-posedness and exponential stability of the linear model

This section is dedicated to establishing the well-posedness of the linear model, which will be derived based on a theorem from the literature. The analysis aims to rigorously confirm the existence, uniqueness, and continuous dependence of solutions on initial data. Additionally, we delve into the exponential decay of the model, extending the discussion to scenarios where more relaxed conditions are imposed on the coefficient of the time-delayed term, highlighting the robustness of the model under less restrictive assumptions. First we consider the linear model

$$\begin{cases} u_t(t, x) + u_{xxx}(t, x) - u_{xx}(t, x) + h(x)u(t, x) + \lambda(x)u(t - \tau, x) = 0, & \text{in } \mathbb{R} \times (0, \infty), \\ u(x, s) = u_0(x, s), & \text{in } \mathbb{R} \times [-\tau, 0], \end{cases} \quad (2.1)$$

and the undelayed version

$$\begin{cases} u_t(t, x) + u_{xxx}(t, x) - u_{xx}(t, x) + h(x)u(t, x) = 0, & \text{in } \mathbb{R} \times (0, \infty), \\ u(x, 0) = u_0(x, 0), & \text{in } \mathbb{R}. \end{cases} \quad (2.2)$$

For simplicity of notation we denote $u^0(x) = u_0(x, 0)$.

For the stability result of Section 2, and for well-posedness and stability result of Section 3, we will use the following assumption. A broader framework will be investigated in Section 4.

$$\exists \alpha_0 > 0 \quad \text{such that} \quad h(x) \geq \alpha_0 \quad \text{for a.e. } x \in \mathbb{R}. \quad (2.3)$$

The following result, proven in [11], guarantees that problem (2.2) is well-posed in $L^2(\mathbb{R})$ when h belongs to $L^\infty(\mathbb{R})$.

Proposition 2.1. *If $h \in L^\infty(\mathbb{R})$, then the operator $\mathcal{A}_h u := -u_{xxx} + u_{xx} - hu$ on $D(\mathcal{A}_h) := H^3(\mathbb{R})$ generates a strongly continuous semigroup. In addition, if h satisfies (2.3), then $\mathcal{A}_h$ generates a strongly continuous semigroup of contractions in the Hilbert space $L^2(\mathbb{R})$.*

By employing the iterative procedure outlined in [34] alongside standard semigroup arguments from [36], we derive the following well-posedness result for the linear delayed model (2.1).

Theorem 2.2. *If $h, \lambda \in L^\infty(\mathbb{R})$ and $u_0 \in C([-\tau, 0]; L^2(\mathbb{R}))$, then there exists a unique solution $u \in C([-\tau, +\infty); L^2(\mathbb{R}))$ of the problem (2.1).*

Before presenting the result on the exponential decay rate, assuming $\lambda \in L^\infty(\mathbb{R})$, we introduce the following Lyapunov functionals:

$$E(t) := E(u(t)) = \frac{1}{2} \int_{\mathbb{R}} u^2(x, t) dx + \frac{1}{2} \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds, \quad (2.4)$$

and

$$\mathcal{E}(t) = E(t) + c_0 F(t), \quad (2.5)$$

where

$$F(t) = \int_{t-\tau}^t \int_s^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, \sigma) dx d\sigma ds, \quad (2.6)$$

and c_0 is a positive constant to be determined later on.

Here, we examine the case where the damping coefficient is bounded below by a positive constant. Additionally, we impose less restrictive conditions on the coefficient of the delayed feedback term compared to those considered in the literature, specifically in [29]. We assume that h satisfies (2.3). Moreover, there exist a constant α and a function $g \in L^p(\mathbb{R})$, for $1 \leq p < \infty$, such that

$$|\lambda(x)| \leq \alpha + g(x) \quad \text{for a.e. } x \in \mathbb{R}, \quad (2.7)$$

with

$$0 \leq \alpha < \alpha_0 \quad \text{and} \quad \|g\|_p < \left(\frac{\alpha_0 - \alpha}{c(p)} \right)^{1 - \frac{1}{2p}}, \quad (2.8)$$

where

$$c(p) := \left(1 - \frac{1}{2p} \right) \left(\frac{2}{p} \right)^{\frac{1}{2p-1}}, \quad \text{for some } 1 \leq p < \infty. \quad (2.9)$$

Remark 2.3. Note that our above conditions significantly improve upon those in [29]. Indeed, we assume (2.7), which is more general than the key assumption in [29]:

$$\frac{e^\tau + 1}{2} |\lambda(x)| \leq \alpha + \beta(x), \quad \text{for a.e. } x \in \mathbb{R}, \quad (2.10)$$

where α is a suitable positive constant and $\beta \in L^p(\mathbb{R})$ has to satisfy some conditions. In particular, our assumption is independent of the time delay, whereas (2.10), for fixed damping coefficients, can be regarded as a smallness restriction on the time delay size. In addition, we emphasize that the particular choice of the Lyapunov functionals (2.4)–(2.6) plays a crucial role in deriving the improved assumption given in (2.7).

The following exponential decay result holds for solutions of (2.1).

Theorem 2.4. Assume $h, \lambda \in L^\infty(\mathbb{R})$ and h satisfies (2.3). Moreover, assume that there exist a positive constant α and a function $g \in L^p(\mathbb{R})$, for some $1 \leq p < \infty$, such that λ satisfies (2.7) with (2.8). Then, there exists a constant $c_0 > 0$ such that the solution u of the problem (2.1) satisfies the exponential decay estimate

$$E(t) \leq C(u^0, c_0) e^{-\delta_1(c_0)t}, \quad (2.11)$$

for a suitable constant $\delta_1(c_0) > 0$, with

$$C(u^0, c_0) = \frac{1}{2} \|u^0\|_2^2 + \frac{1}{2} \int_{-\tau}^0 \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds + c_0 \int_{-\tau}^0 \int_s^0 \int_{\mathbb{R}} |\lambda(x)| u^2(x, \sigma) dx d\sigma ds. \quad (2.12)$$

Proof. By differentiating $\mathcal{E}(t)$ and using (2.1), we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &= \int_{\mathbb{R}} uu_t dx + \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx - \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t - \tau) dx \\ &\quad - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds + c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx ds \\ &= - \int_{\mathbb{R}} uu_{xxx} dx + \int_{\mathbb{R}} uu_{xx} dx - \int_{\mathbb{R}} h(x) u^2(x, t) dx - \int_{\mathbb{R}} \lambda(x) u(x, t) u(x, t - \tau) dx \\ &\quad + \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx - \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t - \tau) dx \\ &\quad - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds + c_0 \tau \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx. \end{aligned}$$

Integrating by parts, using the fact that $\int_{\mathbb{R}} u_{xxx} u dx = 0$, $\forall u \in H^3(\mathbb{R})$, and using Young's inequality, we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - \int_{\mathbb{R}} h(x) u^2(x, t) dx + \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx \\ &\quad + \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t - \tau) dx + \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx - \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t - \tau) dx \\ &\quad - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds + c_0 \tau \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx. \end{aligned}$$

Now, using the assumptions (2.3) and (2.7) in the above equation, we infer

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - \alpha_0 \int_{\mathbb{R}} u^2(x, t) dx + (1 + c_0 \tau) \alpha \int_{\mathbb{R}} u^2(x, t) dx \\ &\quad + (1 + c_0 \tau) \int_{\mathbb{R}} g(x) u^2(x, t) dx - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx. \end{aligned}$$

Applying Hölder inequality, we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - (\alpha_0 - (1 + c_0 \tau) \alpha) \int_{\mathbb{R}} u^2(x, t) dx \\ &\quad + (1 + c_0 \tau) \left(\int_{\mathbb{R}} |g(x)|^p dx \right)^{\frac{1}{p}} \left(\int_{\mathbb{R}} |u(x, t)|^{2q} dx \right)^{\frac{1}{q}} - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx, \end{aligned} \quad (2.13)$$

with $q = \frac{p}{p-1}$. By noting that (as discussed in [11]), for all $u \in H^1(\mathbb{R})$, we have $\|u\|_\infty^2 \leq 2\|u\|_2\|u_x\|_2$. Then, we can infer that

$$\begin{aligned} \left(\int_{\mathbb{R}} |u(x, t)|^{2q} dx \right)^{\frac{1}{q}} &= \left(\int_{\mathbb{R}} u(x, t)^2 (u(x, t))^{\frac{2}{p-1}} dx \right)^{\frac{1}{q}} \leq \|u\|_2^{\frac{2}{q}} \|u\|_\infty^{\frac{2}{q(p-1)}} = \|u\|_2^{\frac{2}{q}} \|u\|_\infty^{\frac{2}{p}} \\ &\leq 2^{\frac{1}{p}} \|u\|_2^{\frac{2}{q}} \|u\|_2^{\frac{1}{p}} \|u_x\|_2^{\frac{1}{p}} = 2^{\frac{1}{p}} \|u\|_2^{\frac{2p-1}{p}} \|u_x\|_2^{\frac{1}{p}}. \end{aligned}$$

Using the above inequality in (2.13) and using the fact that $q = \frac{p}{p-1}$, we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - (\alpha_0 - (1 + c_0 \tau) \alpha) \int_{\mathbb{R}} u^2(x, t) dx \\ &\quad + 2^{\frac{1}{p}} (1 + c_0 \tau) \|g\|_p \|u\|_2^{\frac{2p-1}{p}} \|u_x\|_2^{\frac{1}{p}} - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx. \end{aligned} \quad (2.14)$$

Now, using Young's inequality in the third term of the right hand side of the above inequality, and by fixing $\varepsilon > 0$, we obtain

$$\begin{aligned} 2^{\frac{1}{p}} (1 + c_0 \tau) \|g\|_p \|u\|_2^{\frac{2p-1}{p}} \|u_x\|_2^{\frac{1}{p}} &\leq \left(\frac{1}{\varepsilon} (1 + c_0 \tau) \|g\|_p \|u\|_2^{\frac{2p-1}{p}} \right) \left(\varepsilon 2^{\frac{1}{p}} \|u_x\|_2^{\frac{1}{p}} \right) \\ &\leq \frac{\left(\frac{1}{\varepsilon} (1 + c_0 \tau) \|g\|_p \|u\|_2^{\frac{2p-1}{p}} \right)^{2p}}{\frac{2p}{2p-1}} + \frac{\left(\varepsilon 2^{\frac{1}{p}} \|u_x\|_2^{\frac{1}{p}} \right)^{2p}}{2p}. \end{aligned}$$

By using the above inequality in (2.14) and by choosing ε such that $4\varepsilon^{2p} = 2p$, we can infer

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - (\alpha_0 - (1 + c_0 \tau) \alpha) \int_{\mathbb{R}} u^2(x, t) dx \\ &\quad + \left(\frac{2p-1}{2p} \right) (1 + c_0 \tau)^{\frac{2p}{2p-1}} \left(\frac{2}{p} \right)^{\frac{1}{2p-1}} \|g\|_p^{\frac{2p}{2p-1}} \|u\|_2^2 + \|u_x\|_2^2 - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx. \end{aligned}$$

Thus, we can conclude that

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \left(\alpha_0 - (1 + c_0 \tau) \alpha - c(p)(1 + c_0 \tau)^{\frac{2p}{2p-1}} \|g\|_p^{\frac{2p}{2p-1}} \right) \int_{\mathbb{R}} u^2(x, t) dx \\ &\quad - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx. \end{aligned} \quad (2.15)$$

From (2.8), we can deduce that for a sufficiently small c_0 , the following inequalities hold:

$$0 \leq (1 + c_0 \tau) \alpha < \alpha_0 \quad \text{and} \quad \|g\|_p < \left(\frac{\alpha_0 - (1 + c_0 \tau) \alpha}{c(p)(1 + c_0 \tau)^{\frac{2p}{2p-1}}} \right)^{1 - \frac{1}{2p}}. \quad (2.16)$$

Then, from (2.16), we have that

$$\alpha_0 - (1 + c_0 \tau) \alpha - c(p)(1 + c_0 \tau)^{\frac{2p}{2p-1}} \|g\|_p^{\frac{2p}{2p-1}} > 0.$$

Therefore, inequalities (2.15) and (2.16) yield to

$$\frac{d}{dt} \mathcal{E}(t) \leq -\delta_0(c_0) \mathcal{E}(t), \quad (2.17)$$

$$\text{where } \delta_0(c_0) = \min \left\{ 2 \left(\alpha_0 - (1 + c_0 \tau) \alpha - c(p)(1 + c_0 \tau)^{\frac{2p}{2p-1}} \|g\|_p^{\frac{2p}{2p-1}} \right), 2c_0 \right\}.$$

Furthermore, we can observe that

$$\begin{aligned} \mathcal{E}(t) &= E(t) + c_0 \int_{t-\tau}^t \int_{\mathbb{R}} \int_{\mathbb{R}} |\lambda(x)| u^2(x, \sigma) dx d\sigma ds \\ &= E(t) + c_0 \int_{t-\tau}^t \int_{t-\tau}^s \int_{\mathbb{R}} |\lambda(x)| u^2(x, \sigma) dx d\sigma ds \\ &\leq E(t) + c_0 \tau \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, \sigma) dx d\sigma \\ &\leq \frac{1}{2} \int_{\mathbb{R}} u^2(x, t) dx + \frac{1}{2} (1 + 2c_0 \tau) \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, \sigma) dx d\sigma \\ &\leq (1 + 2c_0 \tau) E(t). \end{aligned} \quad (2.18)$$

Thus, using (2.18) in (2.17) yields to

$$\frac{d}{dt} \mathcal{E}(t) \leq -\delta_1(c_0) \mathcal{E}(t). \quad (2.19)$$

where $\delta_1(c_0) = \frac{\delta_0(c_0)}{1 + 2c_0 \tau}$. Thus, using Gronwall's Lemma and the fact that $E(t) \leq \mathcal{E}(t)$, we obtain (2.11) with $C(u^0, c_0) = \mathcal{E}(0)$. $\square$

Remark 2.5. From the proof above we see that the constant $c_0 > 0$ in the statement of Theorem 2.4 has to be small enough to satisfy (2.16). This allows us to obtain the exponential decay estimate (2.11) with rate, as detailed in the proof,

$$\delta_1(c_0) = \frac{2}{1 + 2c_0 \tau} \min \left\{ \alpha_0 - (1 + c_0 \tau) \alpha - c(p)(1 + c_0 \tau)^{\frac{2p}{2p-1}} \|g\|_p^{\frac{2p}{2p-1}}, c_0 \right\}. \quad (2.20)$$

Remark 2.6. We present here an example of the selection of c_0 :

- If $\|g\|_p \neq 0$:

$$c_0 = \frac{1}{2\tau} \min \left\{ \frac{\sqrt{\alpha^2 + 4\alpha_0 c(p) \|g\|_p^{\frac{2p}{2p-1}}} - \alpha}{2c(p) \|g\|_p^{\frac{2p}{2p-1}}} - 1, \frac{\alpha_0}{\alpha} - 1 \right\}. \quad (2.21)$$

First, we need to verify that $c_0 > 0$. Note that $\alpha < \alpha_0$ implies

$$\frac{\alpha_0}{\alpha} - 1 > 0. \quad (2.22)$$

Moreover, from the condition (2.8), we have that

$$\alpha_0 > \alpha + c(p) \|g\|_p^{\frac{2p}{2p-1}}. \quad (2.23)$$

Multiplying both sides of (2.23) by $4c(p) \|g\|_p^{\frac{2p}{2p-1}}$ and adding α^2 , we obtain

$$\alpha^2 + 4\alpha_0 c(p) \|g\|_p^{\frac{2p}{2p-1}} > \alpha^2 + 4\alpha c(p) \|g\|_p^{\frac{2p}{2p-1}} + 4c^2(p) \|g\|_p^{\frac{4p}{2p-1}},$$

which can be equivalently written as

$$\alpha^2 + 4\alpha_0 c(p) \|g\|_p^{\frac{2p}{2p-1}} > \left(\alpha + 2\alpha c(p) \|g\|_p^{\frac{2p}{2p-1}} \right)^2.$$

Thus, we deduce that

$$\frac{\sqrt{\alpha^2 + 4\alpha_0 c(p) \|g\|_p^{\frac{2p}{2p-1}}} - \alpha}{2c(p) \|g\|_p^{\frac{2p}{2p-1}}} > 1,$$

this, together with (2.22), ensures that $c_0 > 0$.

Now, we need to verify the two conditions in (2.16) are satisfied with c_0 as in (2.21). First, we have

$$c_0 < \frac{1}{\tau} \left(\frac{\alpha_0}{\alpha} - 1 \right),$$

implying directly that

$$(1 + c_0 \tau) \alpha < \alpha_0.$$

Next, for the second condition, from the definition of c_0 , we have

$$c_0 < \frac{1}{\tau} \left(\frac{\sqrt{\alpha^2 + 4\alpha_0 c(p) \|g\|_p^{\frac{2p}{2p-1}}} - \alpha}{2c(p) \|g\|_p^{\frac{2p}{2p-1}}} - 1 \right).$$

This implies that

$$\alpha + 2(1 + c_0 \tau) c(p) \|g\|_p^{\frac{2p}{2p-1}} < \sqrt{\alpha^2 + 4\alpha_0 c(p) \|g\|_p^{\frac{2p}{2p-1}}},$$

which gives

$$c(p)(1 + c_0 \tau)^2 \|g\|_p^{\frac{2p}{2p-1}} < \alpha_0 - \alpha(1 + c_0 \tau). \quad (2.24)$$

Since $p \geq 1$, we have $\frac{2p}{2p-1} \leq 2$, and so

$$(1 + c_0 \tau)^{\frac{2p}{2p-1}} \leq (1 + c_0 \tau)^2.$$

Using the above inequality in (2.24), yields to

$$c(p)(1 + c_0 \tau)^{\frac{2p}{2p-1}} \|g\|_p^{\frac{2p}{2p-1}} < \alpha_0 - \alpha(1 + c_0 \tau),$$

which implies

$$\|g\|_p^{\frac{2p}{2p-1}} < \frac{\alpha_0 - (1 + c_0 \tau) \alpha}{c(p)(1 + c_0 \tau)^{\frac{2p}{2p-1}}},$$

and hence the second condition is verified.

- If $\|g\|_p = 0$: This implies that $g = 0$ a.e.

From the computations in [Theorem 2.4](#), in particular [\(2.15\)](#), we deduce that only one condition needs to be satisfied:

$$(1 + c_0\tau)\alpha < \alpha_0.$$

Consequently, an example of c_0 in this case is any number satisfying

$$c_0 < \frac{1}{\tau} \left(\frac{\alpha_0}{\alpha} - 1 \right).$$

3. Well-posedness and exponential decay for the nonlinear model

In order to prove the well-posedness of the nonlinear model [\(1.4\)](#) we consider first the corresponding linear inhomogeneous initial value problem

$$\begin{cases} u_t(t, x) + u_{xxx}(t, x) - u_{xx}(t, x) + h(x)u(t, x) \\ \quad + \lambda(x)u(t - \tau, x) = f(x, t) & \text{in } \mathbb{R} \times (0, T), \\ u(x, s) = u_0(x, s), & \text{in } \mathbb{R} \times [-\tau, 0], \end{cases} \quad (3.1)$$

for some $0 < T < \infty$. Using the operator $\mathcal{A}_h$ defined in [Proposition 2.1](#), [\(3.1\)](#) can be rewritten in the form

$$\begin{cases} u_t(t, x) + \lambda(x)u(t - \tau, x) = \mathcal{A}_h u(x, t) + f(x, t) & \text{in } \mathbb{R} \times (0, \infty), \\ u(x, s) = u_0(x, s) & \text{in } [-\tau, 0]. \end{cases} \quad (3.2)$$

In the whole section we assume [\(2.3\)](#). A more general framework will be studied later on. According to [Proposition 2.1](#), $\mathcal{A}_h$ generates a strongly continuous semigroup of contractions $(S(t))_{t \geq 0}$ in $L^2(\mathbb{R})$. Then, for any initial data $u_0 \in C([-\tau, 0], L^2(\mathbb{R}))$ and $f \in L^1(0, T; L^2(\mathbb{R}))$, [\(3.2\)](#) has a unique mild solution $u \in C([-\tau, T], L^2(\mathbb{R}))$, given by the following formula:

$$u(t) = S(t)u^0 - \int_0^t S(t - \tau)\lambda u(s - \tau)ds + \int_0^t S(t - s)f(s)ds, \quad t \in [0, T]. \quad (3.3)$$

As proven in [\[29\]](#), this mild solution depends continuously on the data:

$$\|u\|_{C([0, T], L^2(\mathbb{R}))} := \sup_{[0, T]} \|u\|_{L^2(\mathbb{R})} \leq e^{\|\lambda\|_\infty T} \left(\|u^0\|_2 + \|f\|_{L^1(0, T; L^2(\mathbb{R}))} + \|\lambda\|_\infty \int_{-\tau}^0 \|u(s)\|_2 ds \right).$$

Moreover, the solution of [\(3.2\)](#) has an additional space regularity. Let us introduce the Banach space

$$\mathcal{B}_T := C([0, T]; L^2(\mathbb{R})) \cap L^2(0, T; H^1(\mathbb{R})),$$

with the following norm

$$\|u\|_{\mathcal{B}_T} := \|u\|_{C([0, T]; L^2(\mathbb{R}))} + \|\partial_x u\|_{L^2(0, T; H^1(\mathbb{R}))}.$$

We have the following proposition which is proved in [\[29\]](#).

Proposition 3.1. *If $u_0 \in C([-\tau, 0], L^2(\mathbb{R}))$ and $f \in L^1(0, T; L^2(\mathbb{R}))$, then the solution of [\(3.2\)](#) belongs to $\mathcal{B}_T$ and*

$$\|u\|_{\mathcal{B}_T} \leq C_T \left(\|u^0\|_2 + \|f\|_{L^1(0, T; L^2(\mathbb{R}))} + (\|\lambda\|_\infty \tau^{\frac{1}{2}} + \|\lambda\|_\infty^{\frac{1}{2}}) \|u\|_{L^2(-\tau, 0; L^2(\mathbb{R}))} \right), \quad (3.4)$$

where $C_T = \sqrt{\frac{3}{2}} (1 + e^{2\|\lambda\|_\infty T})^{\frac{1}{2}} e^{(\|\lambda\|_\infty + \|h\|_\infty)T}$. Moreover, we have that the following identity holds for all $t \in [0, T]$:

$$\begin{aligned} \|u(t)\|_2^2 + 2 \int_0^t \|u_x(s)\|_2^2 ds + 2 \int_0^t \int_{\mathbb{R}} h u^2(x, s) dx ds \\ + 2 \int_0^t \int_{\mathbb{R}} \lambda u(x, s - \tau) u(x, s) dx ds = \|u_0\|_2^2 + 2 \int_0^t \int_{\mathbb{R}} f(x, s) u(x, s) dx ds. \end{aligned} \quad (3.5)$$

Prior to establishing the well-posedness of the nonlinear model [\(1.4\)](#), we shall present some technical lemmas that will serve as foundational tools in the proofs to follow. For a proof of the following two lemmas we refer to [\[19\]](#).

Lemma 3.2. *Let $a \in C^0(\mathbb{R})$ be a function satisfying*

$$|a(\xi)| \leq C(1 + |\xi|^r), \quad \forall \xi \in \mathbb{R}, \quad (3.6)$$

with $0 \leq r < 2$. Then, there exists a positive constant C such that for any $T > 0$ and $u, v \in \mathcal{B}_T$, we have

$$\|a(u)v_x\|_{L^1(0, T; L^2(\mathbb{R}))} \leq 2^{\frac{r}{2}} C T^{\frac{2-r}{4}} \|u\|_{\mathcal{B}_T}^r \|v\|_{\mathcal{B}_T} + C T^{\frac{1}{2}} \|v\|_{\mathcal{B}_T}. \quad (3.7)$$

Lemma 3.3. *Let $h \in L^\infty(\mathbb{R})$ and $u, v, w \in \mathcal{B}_T$. Then, we have the following*

$$(1) \quad \|hu\|_{L^1(0, T; L^2(\mathbb{R}))} \leq T^{\frac{1}{2}} \|h\|_\infty \|u\|_{\mathcal{B}_T}.$$

$$(2) \quad \|uw_x\|_{L^1(0, T; L^2(\mathbb{R}))} \leq 2^{\frac{1}{2}} T^{\frac{1}{4}} \|u\|_{\mathcal{B}_T} \|w\|_{\mathcal{B}_T}.$$

(3) If $1 \leq r < 2$, then $\|u|v|^{r-1}w_x\|_{L^1(0,T;L^2(\mathbb{R}))} \leq 2^{\frac{r}{2}}T^{\frac{2-r}{4}}\|u\|_{B_T}\|w\|_{B_T}\|v\|_{B_T}^{r-1}$.

(4) If $a \in C^1(\mathbb{R})$ is a function satisfying

$$|a(\xi)| \leq C(1 + |\xi|^r) \quad \text{and} \quad |a'(\xi)| \leq C(1 + |\xi|^{r-1}), \quad \forall \xi \in \mathbb{R}, \quad (3.8)$$

with $1 \leq r < 2$, then the map $\mathcal{M} : B_T \rightarrow L^1(0, T; L^2(\mathbb{R}))$ defined by $\mathcal{M}u := a(u)u_x$ is locally Lipschitz continuous and we have

$$\begin{aligned} \|\mathcal{M}u - \mathcal{M}v\|_{L^1(0,T;L^2(\mathbb{R}))} \\ \leq C\left(\sqrt{2}T^{\frac{1}{4}}\|u\|_{B_T} + 2^{\frac{r}{2}}T^{\frac{2-r}{4}}\left(\|u\|_{B_T}^r + \|u\|_{B_T}\|v\|_{B_T}^{r-1} + \|v\|_{B_T}^r\right) + T^{\frac{1}{2}}\right)\|u - v\|_{B_T}. \end{aligned} \quad (3.9)$$

Now, we establish a local existence result, which will subsequently allow us to infer the existence of a global mild solution of (1.4).

Proposition 3.4. Let $a \in C^1(\mathbb{R})$ be a function satisfying

$$|a(\xi)| \leq C(1 + |\xi|^r) \quad \text{and} \quad |a'(\xi)| \leq C(1 + |\xi|^{r-1}), \quad \forall \xi \in \mathbb{R}, \quad (3.10)$$

with $1 \leq r < 2$. Let $h, \lambda \in L^\infty(\mathbb{R})$ and $u_0 \in C([-\tau, 0]; L^2(\mathbb{R}))$. Then, the problem (1.4) has a unique mild solution on $[0, T)$ for some $T > 0$. Moreover,

$$\begin{aligned} \|u(t)\|_2^2 + 2\int_0^t \|u_x(s)\|_2^2 ds + 2\int_0^t \int_{\mathbb{R}} h(x)u^2(x, s) dx ds \\ + 2\int_0^t \int_{\mathbb{R}} \lambda(x)u(x, s - \tau)u(x, s) dx ds = \|u^0\|_2^2, \quad \forall t \in [0, T). \end{aligned} \quad (3.11)$$

Proof. Since $\mathcal{A}_h$ generates a strongly continuous semigroup of contractions $(S(t))_{t \geq 0}$, and using Proposition 3.1, the solution of (3.2) satisfies the estimate (3.4) with $f = -a(u)u_x$. Then,

$$\|u\|_{B_T} \leq C_T \left(\|u^0\|_2 + \|\mathcal{M}u\|_{L^1(0,T;L^2(\mathbb{R}))} + (\|\lambda\|_\infty \tau^{\frac{1}{2}} + \|\lambda\|_\infty^{\frac{1}{2}}) \|u\|_{L^2(-\tau,0;L^2(\mathbb{R}))} \right), \quad (3.12)$$

with $\mathcal{M}u = a(u)u_x$.

Let $u_0 \in C([-\tau, 0]; L^2(\mathbb{R}))$ denote the given initial datum. Define the mapping Λ on the Banach space B_T , equipped with its natural norm, as:

$$(\Lambda u)(t) = S(t)u^0 - \int_0^t S(t-s)\lambda u_0(s-\tau) ds - \int_0^t S(t-s)a(u)u_x(s) ds, \quad t \in [0, \tau]. \quad (3.13)$$

For $T \leq \tau$, by using Proposition 3.1, it follows that for any $v, w \in B_T$, we have

$$\|\Lambda v - \Lambda w\|_{B_T} \leq C_T \|\mathcal{M}v - \mathcal{M}w\|_{L^1(0,T;L^2(\mathbb{R}))}. \quad (3.14)$$

Applying part (4) of Lemma 3.3, we obtain the estimate:

$$\begin{aligned} \|\Lambda v - \Lambda w\|_{B_T} \\ \leq C_T C \left(\sqrt{2}T^{\frac{1}{4}}\|v\|_{B_T} + 2^{\frac{r}{2}}T^{\frac{2-r}{4}}\left(\|v\|_{B_T}^r + \|v\|_{B_T}\|w\|_{B_T}^{r-1} + \|w\|_{B_T}^r\right) + T^{\frac{1}{2}} \right) \|v - w\|_{B_T}, \end{aligned} \quad (3.15)$$

for all $0 < T \leq \tau$.

Now, consider the closed ball $B_R(0)$ in the Banach space B_T , defined as:

$$B_R(0) := \left\{ u \in B_T : \|u\|_{B_T} \leq R \right\}. \quad (3.16)$$

For any $v, w \in B_R(0)$, using (3.15), we deduce

$$\|\Lambda v - \Lambda w\|_{B_T} \leq C_T C \left(\sqrt{2}T^{\frac{1}{4}}R + 2^{\frac{r}{2}}T^{\frac{2-r}{4}}3R^r + T^{\frac{1}{2}} \right) \|v - w\|_{B_T}. \quad (3.17)$$

Thus, we can write:

$$\|\Lambda v - \Lambda w\|_{B_T} \leq C_1(T) \|v - w\|_{B_T}, \quad (3.18)$$

where $C_1(T) = C_T C \left(\sqrt{2}T^{\frac{1}{4}}R + 2^{\frac{r}{2}}T^{\frac{2-r}{4}}3R^r + T^{\frac{1}{2}} \right)$.

Now, fix $R := 2C(\tau, \tau)(\|u^0\|_2 + \|u_0\|_{L^2(-\tau,0;L^2(\mathbb{R}))})$, where

$$C(T, \tau) = C_T \max \left\{ 1, \|\lambda\|_\infty \tau^{\frac{1}{2}} + \|\lambda\|_\infty^{\frac{1}{2}} \right\}. \quad (3.19)$$

Accordingly, choose $T > 0$ small enough such that $C_1(T) < \frac{1}{2}$. Next, setting $w \equiv 0$ in (3.15), we have:

$$\begin{aligned} \left\| \Lambda v - S(t)u_0(0) + \int_0^t S(t-s)\lambda u_0(s-\tau) ds \right\|_{B_T} \\ \leq C_T C \left(\sqrt{2}T^{\frac{1}{4}}\|v\|_{B_T} + 2^{\frac{r}{2}}T^{\frac{2-r}{4}}\|v\|_{B_T}^r + T^{\frac{1}{2}} \right) \|v\|_{B_T}. \end{aligned} \quad (3.20)$$

From this and Proposition 3.1, we deduce

$$\|\Lambda v\|_{B_T} \leq C_T C \left(\sqrt{2} T^{\frac{1}{4}} \|v\|_{B_T} + 2^{\frac{r}{2}} T^{\frac{2-r}{4}} \|v\|_{B_T}^r + T^{\frac{1}{2}} \right) \|v\|_{B_T} + C(T, \tau) (\|u^0\|_2 + \|u_0\|_{L^2(-\tau, 0; L^2(\mathbb{R}))}),$$

where $C(T, \tau)$ is defined in (3.19).

Assuming $v \in B_R(0)$ and being $T \leq \tau$, it follows that

$$\|\Lambda v\|_{B_T} \leq C_2(T) R + C(T, \tau) (\|u^0\|_2 + \|u_0\|_{L^2(-\tau, 0; L^2(\mathbb{R}))}),$$

where $C_2(T) = C_T C \left(\sqrt{2} T^{\frac{1}{4}} R + 2^{\frac{r}{2}} T^{\frac{2-r}{4}} R^r + T^{\frac{1}{2}} \right)$.

Since $C_2(T) \leq C_1(T) < \frac{1}{2}$, and recalling the definition of R , we conclude:

$$\|\Lambda v\|_{B_T} \leq R. \quad (3.21)$$

From (3.18) and (3.21), it follows that $\Lambda : B_R(0) \rightarrow B_R(0)$ is a contraction mapping. By the Banach fixed-point theorem, there exists a unique $u \in B_R(0)$ such that $\Lambda u = u$. Thus, u is the unique local mild solution of (1.4), proving the local well-posedness for $0 < T \leq \tau$ small enough.

Now, multiplying (1.4) by u , we obtain

$$\begin{aligned} & \int_0^t \int_{\mathbb{R}} u_s(x, s) u(x, s) dx ds + \int_0^t \int_{\mathbb{R}} u_{xxx}(x, s) u(x, s) dx ds \\ & - \int_0^t \int_{\mathbb{R}} u_{xx}(x, s) u(x, s) dx ds + \int_0^t \int_{\mathbb{R}} h u^2(x, s) dx ds \\ & + \int_0^t \int_{\mathbb{R}} \lambda u(x-s, \tau) u(x, s) dx ds + \int_0^t \int_{\mathbb{R}} a(u) u_x(x, s) u(x, s) dx ds = 0. \end{aligned}$$

By using the fact that $\int_{\mathbb{R}} u_{xxx} u dx = 0$, $\forall u \in H^3(\mathbb{R})$, and using that

$$\int_0^t \int_{\mathbb{R}} a(u) u_x(x, s) u(x, s) dx ds = 0, \quad (3.22)$$

since

$$\int_{\mathbb{R}} a(u) u_x(x, s) u(x, s) dx = \int_{\mathbb{R}} \left(\int_0^{u(x)} v a(v) dv \right)_x dx = 0, \quad (3.23)$$

after integration by parts, we can infer

$$\begin{aligned} & \frac{1}{2} \int_{\mathbb{R}} u^2(x, t) dx - \frac{1}{2} \int_{\mathbb{R}} u^2(x, 0) dx + \int_0^t \int_{\mathbb{R}} u_x^2(x, s) dx ds \\ & + \int_0^t \int_{\mathbb{R}} h u^2(x, s) dx ds + \int_0^t \int_{\mathbb{R}} \lambda u(x-s, \tau) u(x, s) dx ds = 0, \end{aligned}$$

and thus we deduce that (3.11) is satisfied. $\square$

Theorem 3.5. Let $h, \lambda \in L^\infty(\mathbb{R})$ and $u_0 \in C([-\tau, 0]; L^2(\mathbb{R}))$ such that h satisfies (2.3) and λ satisfies (2.7) and (2.8). Let $a \in C^1(\mathbb{R})$ be a function satisfying (3.10) with $1 \leq r < 2$. Then, there exists a global mild solution of (1.4). Moreover, for all $t \geq 0$, the following energy identity is satisfied:

$$\begin{aligned} & \|u(t)\|_2^2 + 2 \int_0^t \|u_x(s)\|_2^2 ds + 2 \int_0^t \int_{\mathbb{R}} h(x) u^2(x, s) dx ds \\ & + 2 \int_0^t \int_{\mathbb{R}} \lambda(x) u(x, s-\tau) u(x, s) dx ds = \|u_0\|_2^2. \end{aligned} \quad (3.24)$$

Proof. Let $[0, T)$ be the existence time interval for the solution u of (1.4). By multiplying the equation in (1.4) by u and integrating by parts, we derive (3.24). Differentiating $\mathcal{E}(t)$ with respect to t and proceeding with similar computations to that in Theorem 2.4, while assuming (2.3)–(2.8), we deduce that

$$\frac{d}{dt} \mathcal{E}(t) \leq 0. \quad (3.25)$$

This ensures that $\|u\|_2$ remains bounded for $t \in [0, T)$. Combining this with (3.24), we conclude that $\|u\|_{B_T}$ remains bounded for $t \in [0, T)$. Consequently, the solution can be extended to any time interval, yielding a global mild solution $u \in C([0, +\infty); L^2(\mathbb{R})) \cap L^2((0, +\infty); H^1(\mathbb{R}))$. $\square$

Theorem 3.6. Let $h, \lambda \in L^\infty(\mathbb{R})$ and $u_0 \in C([-\tau, 0]; L^2(\mathbb{R}))$ such that h satisfies (2.3) and λ satisfies (2.7) with (2.8). Let $a \in C^1(\mathbb{R})$ be a function satisfying (3.10) with $1 \leq r \leq 2$. Then, the model (1.4) is exponentially stable. In particular, the solution of (1.4) satisfies the following estimation

$$E(t) \leq C(u^0, c_0) e^{-\delta_1(c_0)t}, \quad (3.26)$$

with c_0 and $\delta_1(c_0)$ as in Remark 2.5 and $C(u^0, c_0)$ as in (2.12).

Proof. By differentiating the Lyapunov functional defined in (2.5), and following similar arguments to the ones in the proof of Theorem 2.4, we can obtain, as before,

$$\frac{d\mathcal{E}}{dt} \leq -\delta_1(c_0) \mathcal{E}(t). \quad (3.27)$$

Thus, we can deduce that

$$\mathcal{E}(t) \leq C(u^0, c_0) e^{-\delta_1(c_0)t}, \quad (3.28)$$

and then (4.6) follows. $\square$

4. Exponential decay of the non-linear model in a more general setting

In this section, we explore a broader framework for the damping coefficient, permitting it to change sign, and impose more flexible conditions on the coefficient of the time-delay term. For brevity, we consider only the nonlinear model. The linear case can be studied analogously in the generalized setting.

First, we assume that there exist a positive constant $\alpha_0 > 0$ and a function $g_1 \in L^p(\mathbb{R})$ for $1 \leq p < \infty$, such that

$$h(x) \geq \alpha_0 - g_1(x) \quad \text{for a.e. } x \in \mathbb{R}, \quad (4.1)$$

where the function g_1 satisfies

$$\|g_1\|_p < \left(\frac{\alpha_0}{c(p)} \right)^{1-\frac{1}{2p}}, \quad (4.2)$$

with $c(p)$ defined in (2.9). Moreover, we assume that the function λ satisfies the following

$$|\lambda(x)| \leq \alpha + g(x) \quad \text{for a.e. } x \in \mathbb{R}, \quad (4.3)$$

for some constant α and some function $g \in L^p(\mathbb{R})$ such that

$$0 \leq \alpha < \alpha_0 \quad \text{and} \quad \|g + g_1\|_p < \left(\frac{\alpha_0 - \alpha}{c(p)} \right)^{1-\frac{1}{2p}}. \quad (4.4)$$

We begin by stating the well-posedness theorem, but we omit the proof here since it closely follows the one presented in Theorem 3.5.

Theorem 4.1. *Let $h, \lambda \in L^\infty(\mathbb{R})$ and $u_0 \in C([- \tau, 0]; L^2(\mathbb{R}))$ such that h satisfies (4.1) with (4.2) and λ satisfies (4.3) with (4.4). Let $a \in C^1(\mathbb{R})$ be a function satisfying (3.10), with $1 \leq r < 2$. Then, there exists a global mild solution of (1.4). Moreover, we have that for all $t \geq 0$ the following energy identity is satisfied:*

$$\begin{aligned} \|u(t)\|_2^2 + 2 \int_0^t \|u_x(s)\|_2^2 ds + 2 \int_0^t \int_{\mathbb{R}} h(x) u^2(x, s) dx ds \\ + 2 \int_0^t \int_{\mathbb{R}} \lambda(x) u(x, s - \tau) u(x, s) dx ds = \|u_0\|_2^2. \end{aligned} \quad (4.5)$$

We now present the exponential stability result for the nonlinear model (1.4), under the assumptions (4.1)–(4.4).

Theorem 4.2. *Let $h, \lambda \in L^\infty(\mathbb{R})$ and $u_0 \in C([- \tau, 0]; L^2(\mathbb{R}))$ such that h satisfies (4.1) with (4.2) and λ satisfies (4.3) with (4.4). Let $a \in C^1(\mathbb{R})$ be a function satisfying (3.10) with $1 \leq r < 2$. Then, there exists a sufficiently small constant $c_0 > 0$ such that the solution u of the problem (1.4) satisfies the exponential decay estimate*

$$E(t) \leq C(u^0, c_0) e^{-\delta_2(c_0)t}, \quad (4.6)$$

for a suitable constant $\delta_2(c_0) > 0$, with $C(u^0, c_0)$ as in (2.12).

Proof. We proceed in a similar way as in the proof of Theorem 2.4. By differentiating $\mathcal{E}(t)$, integrating by parts and using Young's inequality, we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - \int_{\mathbb{R}} h(x) u^2(x, t) dx + \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx \\ &\quad + \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t - \tau) dx + \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx - \frac{1}{2} \int_{\mathbb{R}} |\lambda(x)| u^2(x, t - \tau) dx \\ &\quad - c_0 \int_{-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds + c_0 \tau \int_{\mathbb{R}} |\lambda(x)| u^2(x, t) dx. \end{aligned}$$

Now, using the assumptions (4.1) and (4.3) in the above equation, we infer

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - \alpha_0 \int_{\mathbb{R}} u^2(x, t) dx + \int_{\mathbb{R}} g_1(x) u^2(x, t) dx + (1 + c_0 \tau) \alpha \int_{\mathbb{R}} u^2(x, t) dx \\ &\quad + (1 + c_0 \tau) \int_{\mathbb{R}} g(x) u^2(x, t) dx - c_0 \int_{-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds. \end{aligned}$$

Then, we have

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - (\alpha_0 - (1 + c_0 \tau) \alpha) \int_{\mathbb{R}} u^2(x, t) dx \\ &\quad + \int_{\mathbb{R}} (g_1(x) + (1 + c_0 \tau) g(x)) u^2(x, t) dx - c_0 \int_{-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds. \end{aligned}$$

Applying Hölder inequality, we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - (\alpha_0 - (1 + c_0 \tau) \alpha) \int_{\mathbb{R}} u^2(x, t) dx \\ &\quad + \|g_1(x) + (1 + c_0 \tau) g(x)\|_p \|u\|_{2q}^2 - c_0 \int_{-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds, \end{aligned} \quad (4.7)$$

with $q = \frac{p}{p-1}$. Now, observe that (as discussed above), for all $u \in H^1(\mathbb{R})$, we have $\|u\|_\infty^2 \leq 2\|u\|_2 \|u_x\|_2$ and that

$$\|u\|_{2q}^2 \leq \|u\|_2^{\frac{2}{q}} \|u\|_\infty^{\frac{2}{q}} = \|u\|_2^{\frac{2}{q}} \|u\|_\infty^{\frac{2}{q}} \leq 2^{\frac{1}{p}} \|u\|_2^{\frac{2}{q}} \|u\|_2^{\frac{1}{p}} \|u_x\|_2^{\frac{1}{p}} = 2^{\frac{1}{p}} \|u\|_2^{\frac{2p-1}{p}} \|u_x\|_2^{\frac{1}{p}}. \quad (4.8)$$

Then, by using the above inequality in (4.7) and using again the fact that $q = \frac{p}{p-1}$, we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) &\leq - \int_{\mathbb{R}} u_x^2 dx - (\alpha_0 - (1 + c_0 \tau) \alpha) \int_{\mathbb{R}} u^2(x, t) dx \\ &\quad + 2^{\frac{1}{p}} \|g_1(x) + (1 + c_0 \tau) g(x)\|_p \|u\|_2^{\frac{2p-1}{p}} \|u_x\|_2^{\frac{1}{p}} - c_0 \int_{-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds. \end{aligned}$$

Applying Young's inequality, for any fixed $\varepsilon > 0$, we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) \leq & -\|u_x\|_2^2 - (\alpha_0 - (1 + c_0\tau)\alpha) \|u\|_2^2 + \frac{\left(\frac{1}{\varepsilon} \|g_1(x) + (1 + c_0\tau)g(x)\|_p \|u\|_2^{\frac{2p-1}{p}}\right)^{\frac{2p}{2p-1}}}{\frac{2p}{2p-1}} \\ & + \frac{\left(2^{\frac{1}{p}} \varepsilon \|u_x\|_2^{\frac{1}{p}}\right)^{2p}}{2p} - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds. \end{aligned}$$

Finally, by fixing $4\varepsilon^{2p} = 2p$, we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{E}(t) \leq & -\left(\alpha_0 - (1 + c_0\tau)\alpha - c(p) \|g_1(x) + (1 + c_0\tau)g(x)\|_p^{\frac{2p}{2p-1}}\right) \|u\|_2^2 \\ & - c_0 \int_{t-\tau}^t \int_{\mathbb{R}} |\lambda(x)| u^2(x, s) dx ds. \end{aligned} \quad (4.9)$$

From (4.4), we can deduce that for a sufficiently small c_0 , the following inequalities hold:

$$0 \leq (1 + c_0\tau)\alpha < \alpha_0 \quad \text{and} \quad \|g_1 + (1 + c_0\tau)g\|_p < \left(\frac{\alpha_0 - (1 + c_0\tau)\alpha}{c(p)}\right)^{1 - \frac{1}{2p}}. \quad (4.10)$$

Then, from (4.10), we obtain that

$$\alpha_0 - (1 + c_0\tau)\alpha - c(p) \|g_1(x) + (1 + c_0\tau)g(x)\|_p^{\frac{2p}{2p-1}} > 0. \quad (4.11)$$

Therefore, inequalities (4.9) and (4.10) yield to

$$\frac{d}{dt} \mathcal{E}(t) \leq -\delta_2^1(c_0) \mathcal{E}(t), \quad (4.12)$$

where $\delta_2^1(c_0) = 2 \min \left\{ \alpha_0 - (1 + c_0\tau)\alpha - c(p) \|g_1(x) + (1 + c_0\tau)g(x)\|_p^{\frac{2p}{2p-1}}, c_0 \right\}$. By using the inequality derived in (2.18) in the preceding inequality, we arrive at

$$\frac{d}{dt} \mathcal{E}(t) \leq -\delta_2(c_0) \mathcal{E}(t), \quad (4.13)$$

where $\delta_2(c_0) = \frac{\delta_2^1(c_0)}{1 + 2c_0\tau}$. Thus, using Gronwall's Lemma and the fact that $E(t) \leq \mathcal{E}(t)$, we obtain (4.6) with $C(u^0, c_0) = \mathcal{E}(0)$. $\square$

Remark 4.3. From the proof above, we conclude that the constant $c_0 > 0$ in the statement of Theorem 4.2 must be sufficiently small to ensure that conditions (4.10) hold. This constraint enables us to derive the exponential decay estimate (4.6) with the rate, as detailed in the proof,

$$\delta_2(c_0) = \frac{2}{1 + 2c_0\tau} \min \left\{ \alpha_0 - (1 + c_0\tau)\alpha - c(p) \|g_1(x) + (1 + c_0\tau)g(x)\|_p^{\frac{2p}{2p-1}}, c_0 \right\}. \quad (4.14)$$

Remark 4.4. Here, we provide examples of c_0 by distinguishing all possible cases:

- If $\|g_1 + g\|_p \neq 0$ and $\|g\|_p \neq 0$:

A possible choice for the constant c_0 is given by

$$c_0 = \frac{1}{2\tau} \min \left\{ \left(\frac{1-\rho}{2}\right) \left(\frac{\alpha_0}{\alpha} - 1\right), \frac{1}{\|g\|_p} \left[\left(\frac{1+\rho}{2}\right)^{\frac{2p-1}{2p}} - \rho^{\frac{2p-1}{2p}} \right] \left(\frac{\alpha_0 - \alpha}{c(p)}\right)^{\frac{2p-1}{2p}} \right\}, \quad (4.15)$$

where the constant ρ is defined as

$$\rho = \frac{\|g + g_1\|_p^{\frac{2p}{2p-1}}}{\alpha_0 - \alpha} c(p).$$

Note that assumptions (4.4) ensure that $\rho \in (0, 1)$ and then c_0 is positive. We need to verify that this choice of c_0 satisfies the conditions in (4.10). From this choice of c_0 , we obtain

$$\alpha c_0 \tau < \left(\frac{1-\rho}{2}\right) (\alpha_0 - \alpha).$$

Since $\rho \in (0, 1)$, it follows that

$$(1 + c_0\tau)\alpha < \alpha_0,$$

which confirms that the first condition in (4.10) is satisfied.

For the second condition, using the choice of c_0 and ρ , we derive

$$\begin{aligned} \|g_1 + (1 + c_0\tau)g\|_p &\leq \|g_1 + g\|_p + c_0\tau\|g\|_p \\ &< \rho^{\frac{2p-1}{2p}} \left(\frac{\alpha_0 - \alpha}{c(p)} \right)^{\frac{2p-1}{2p}} + \left[\left(\frac{1 + \rho}{2} \right)^{\frac{2p-1}{2p}} - \rho^{\frac{2p-1}{2p}} \right] \left(\frac{\alpha_0 - \alpha}{c(p)} \right)^{\frac{2p-1}{2p}} \\ &= \left(\frac{1 + \rho}{2} \right)^{\frac{2p-1}{2p}} \left(\frac{\alpha_0 - \alpha}{c(p)} \right)^{\frac{2p-1}{2p}} \\ &= \left(1 - \frac{1 - \rho}{2} \right)^{\frac{2p-1}{2p}} \left(\frac{\alpha_0 - \alpha}{c(p)} \right)^{\frac{2p-1}{2p}}. \end{aligned}$$

Thus,

$$\|g_1 + (1 + c_0\tau)g\|_p < \left(\frac{\alpha_0 - \alpha}{c(p)} - \frac{\alpha_0 - \alpha}{c(p)} \left(\frac{1 - \rho}{2} \right) \right)^{\frac{2p-1}{2p}}. \quad (4.16)$$

From the choice of c_0 , we have

$$c_0\tau\alpha < \left(\frac{1 - \rho}{2} \right)(\alpha_0 - \alpha).$$

By using this inequality in (4.16), we obtain

$$\|g_1 + (1 + c_0\tau)g\|_p < \left(\frac{\alpha_0 - (1 + c_0\tau)\alpha}{c(p)} \right)^{\frac{2p-1}{2p}}.$$

Thus, the second condition is also verified.

- If $\|g_1 + g\|_p \neq 0$ and $\|g\|_p = 0$:

In this case, we have $g = 0$ a.e., which simplifies

$$\|g_1 + (1 + c_0\tau)g\|_p = \|g_1\|_p.$$

From the computations in Theorem 4.2, particularly in (4.9), we require

$$\alpha_0 - (1 + c_0\tau)\alpha - c(p)\|g_1\|_p^{\frac{2p}{2p-1}} > 0.$$

Note that, by (4.4),

$$\alpha_0 - \alpha - c(p)\|g_1\|_p^{\frac{2p}{2p-1}} > 0.$$

Thus, a valid choice for c_0 is any number satisfying

$$0 < c_0 < \frac{1}{\tau} \left(\frac{\alpha_0}{\alpha} - 1 - \frac{c(p)}{\alpha} \|g_1\|_p^{\frac{2p}{2p-1}} \right).$$

- If $\|g_1 + g\|_p = 0$ and $\|g\|_p = 0$: This case implies that $g = 0$ a.e. and $g_1 = 0$ a.e. Again, from the computations in (4.9), we only need to verify that

$$\alpha_0 - (1 + c_0\tau)\alpha > 0.$$

Thus, a valid choice for c_0 is any number satisfying

$$0 < c_0 < \frac{1}{\tau} \left(\frac{\alpha_0}{\alpha} - 1 \right).$$

- If $\|g_1 + g\|_p = 0$ and $\|g\|_p \neq 0$:

A possible choice for c_0 is given by

$$c_0 = \frac{1}{2\tau} \min \left\{ 1, \frac{\alpha_0 - \alpha}{\alpha + c(p)\|g\|_p^{\frac{2p}{2p-1}}} \right\}.$$

From the computations, specifically Eq. (4.9), the required condition for c_0 is

$$\|c_0\tau g\|_p < \left(\frac{\alpha_0 - (1 + c_0\tau)\alpha}{c(p)} \right)^{\frac{2p-1}{2p}}. \quad (4.17)$$

Using the chosen value of c_0 , we obtain

$$c_0\tau(\alpha + c(p)\|g\|_p^{\frac{2p}{2p-1}}) < \alpha_0 - \alpha,$$

which implies

$$(1 + c_0\tau)\alpha < \alpha_0 - c(p)c_0\tau \|g\|_p^{\frac{2p}{2p-1}}.$$

Since $c_0\tau < 1$ and using that $\frac{2p}{2p-1} > 1$, we get

$$(1 + c_0\tau)\alpha < \alpha_0 - c(p)(c_0\tau)^{\frac{2p}{2p-1}} \|g\|_p^{\frac{2p}{2p-1}}. \quad (4.18)$$

Thus, condition (4.17) is verified.

We note that the exponential stability of the linear model (2.1) under more general conditions for the damping coefficient h in (4.1) can be established using a similar approach as in Theorem 4.2.

5. Conclusion and open problems

In this work, we have established a robust framework for analyzing the time-delayed generalized Korteweg-de Vries-Burgers equation. By employing semigroup theory and new Lyapunov functional techniques, we have demonstrated global well-posedness and exponential decay of the energy under relaxed assumptions compared to prior studies. The key contributions include a more comprehensive consideration of the damping coefficient's behavior and refined conditions on the delayed term's coefficient, which significantly extend the applicability of existing results. In particular, for given damping coefficients satisfying suitable assumptions, our results are valid without any restrictions on the time delay size. Despite these contributions, some open problems related to this system remain to be addressed. For example, the consideration of a time varying delay feedback can raise several difficulties as well the analysis in bounded domains or the presence of more general nonlinearities.

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